

EvalAudit Paper Strategy

Framing, Scope, Evidence, and a Three-Week Execution Plan

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This reference consolidates the internship chronology, status slides, repository history, journals, runbooks, the inherited Every Eval Ever paper, the TMLR draft skeleton, the OLMo deployment-match confirmation path, and the subsequent paper-framing discussion. It distinguishes what can be published now from what requires additional empirical work.

Evidence cutoff: repository snapshot through July 14, 2026; interpretive revision July 15, 2026.

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1 Consensus update after independent reconstructions

The independent reconstructions have reached substantive consensus. The paper should use a system-identification and identifiability framing; report the OLMo float32 table as a discovery probe; treat the dense OLMo-2 Hugging Face divergence as an unresolved result whose cause may include both execution-stack sensitivity and a dense-model probe artifact; and use recoverable versus non-identifiable provenance as the conceptual spine. The source-level dtype argument is retained as a conditional, per-parameter inference. Live-store observations are collaborator-verified but remain uncitable until their Git objects, manifests, inputs, and hashes are preserved.

2 Executive recommendation

With approximately 2.5–3 weeks remaining, the best target is neither a pure position paper nor a broad benchmark paper. It is:

A systems-and-measurement paper with a clear position, one deep modern empirical case, one historical forensic case, and an optional narrow current-model demonstration.

The paper must remain publishable if the remaining GPU work does not complete. Its core contribution is EvalAudit and a layered account of benchmark reproduction. Its empirical upgrade is an ordinary-path OLMo confirmation. A Qwen comparison is valuable only after that core result and artifact are secured. Fable-distilled models should be treated as a follow-on unless their data and training provenance are already complete.

Primary decision Secure a correct, auditable manuscript first. Increase model count only after the ordinary-HELM OLMo confirmation, result regeneration, and evidence packaging are complete.

3 The two-paper lineage

The repository contains two scientifically distinct paper lines.

3.1 Inherited EEE paper

The NeurIPS Every Eval Ever (EEE) work uses a normalized representation to detect discrepancies on Pythia-6.9B, Vicuna-7B-v1.3, and Falcon-7B across fourteen HELM benchmarks. Its scientific endpoint is discrepancy detection and forensic inspection. Its appendix explicitly leaves unresolved whether residual differences arise from checkpoint, quantization, precision, tokenizer, or generation-kernel choices.

This is inherited, substantially pre-internship work. It should be cited and used as motivation, not retold as a new contribution.

3.2 Internship follow-on

The pre-existing TMLR skeleton is the natural home for the internship’s distinct question:

How far can execution-provenance reconstruction attribute or close the discrepancies that normalized comparison reveals, and when is the original measurement no longer identifiable?

The clean positioning sentence is:

Paper relationship EEE detects reproducibility gaps; the follow-on paper studies their attribution, possible closure, and identifiability.

4 Recommended thesis and paper identity

4.1 Defensible core thesis

Public LLM benchmark artifacts often identify the evaluation recipe but not the complete execution instrument. Exact run-spec replay and controlled reconstruction of model, tokenizer, precision, engine, and harness provenance can attribute—and sometimes close—apparent reproducibility gaps. When that provenance has been lost or post-hoc transformed, the original measurement may remain non-identifiable.

This thesis is supported even if the ordinary-path OLMo confirmation fails.

4.2 Forward-looking systems motivation

The strongest “why care” framing is not old-model preservation for its own sake. Historical HELM runs provide forensic evidence from which to derive requirements for trustworthy current evaluation.

Auditable evaluation is necessary to determine whether measured improvements in small open models exceed variation introduced by the evaluation substrate itself.

The small-model story should be the application of EvalAudit, not a separate benchmark paper attached to a HELM postmortem.

4.3 Recommended title

EvalAudit: Reconstructing the Execution Substrate of LLM Benchmarks

Possible subtitle:

From HELM reproduction failures to auditable evaluation of current local models

Alternative conceptual title:

A Benchmark Recipe Is Not an Experiment: Attribution and Identifiability in LLM Evaluation

5 Current evidence status

Evidence / capability	Status	Defensible interpretation
Exact public <code>run_spec.json</code> replay	Ready	Implemented and shown to correct visible recipe drift, including the documented BBQ prompt mismatch.
Containerized HELM execution and GPU leasing	Ready	Reusable infrastructure exists; individual family grids still require result-level verification.
OLMo tokenizer and template diagnoses	Ready	Special-token and chat-template behavior can change the actual model stimulus and outputs.
OLMo float32/engine result	Partial	Highest-agreement candidates in an oracle-scored 12-instance IFEval probe; not an ordinary HELM reproduction or proof of the historical deployment.
OLMo Hugging Face dense-model divergence	Partial	Prompt mismatch was rejected for an inspected case; root cause remains unresolved and scientifically relevant.
Classic HELM source archaeology	Ready	Strong evidence of carried-forward artifacts, migrated class paths, and an inferred compatible unreleased source interval.
Classic proxy execution	Partial	A live-store review reports complete RedPajama runs under both proxy eras; artifacts remain collaborator-verified and unpreserved. GPT-J/GPT-NeoX/OPT origin instruments remain non-identifiable from the surviving record.
GPT-OSS result store	Partial	Reports are reported freshly regenerated after reporting fixes, but run-artifact and comparison freshness still require manifest-based acceptance and preservation.
OLMo base-model report store	Blocked	The live-store review reports stale unduplicated local values; regenerate before citing.
Qwen result store	Blocked	Combined workflow exists, but the live-store review reports no completed store; treat as genuinely pending.
Population-level reproducibility rate	Blocked	No frozen denominator and no fully regenerated publication store.
Broad small-model progress claim	Blocked	Requires exact model revisions, fair protocols, substrate sensitivity, and complete benchmark results.

6 Per-parameter identifiability

A central refinement is that an experiment need not be wholly identifiable or non-identifiable. Each latent parameter can have a different status. Any corpus-wide percentage, including the internal planning

estimate that roughly 59% of indexed runs required the classic-era path, must be regenerated from a checked-in manifest with an explicit numerator and denominator before publication.

Parameter	Typical status	Implication for the paper
Run specification	Identifiable	Replay the frozen public artifact exactly.
Client/deployment class	Often identifiable	Distinguish hosted, HF in-process, and vLLM paths.
Dtype	Conditional	Explicit pins are identified; unpinned dtype can only be inferred from loader source plus package-version bounds.
Tokenizer/template behavior	Often recoverable	Preserve revisions and rendered and tokenized prompt hashes.
Model/tokenizer revision	Frequently missing	Mutable aliases prevent exact historical recovery.
Engine/package/kernel versions	Often missing	Require prospective execution capsules; diagnose retrospectively by bounded sweeps.
Batching/cache/topology/hardware	Usually missing historically	Measure robustness rather than promising exact hardware replay.
Harness era	Tagged or proxy	Tagged eras can be containerized; migrated classic artifacts may only admit declared proxy instruments.

Historical proxy execution status. A collaborating live-store review reports end-to-end RedPajama-3B runs under both v0.2.4 and v0.3.0 proxy instruments. Because the release package still lacks those raw stores, manifests, and hashes, classify them as collaborator-verified rather than packaged results. Their scientific role is a comparison between declared proxy instruments, not proof that either proxy reproduces the original producing instrument. GPT-J/GPT-NeoX/OPT remain origin cases that are non-identifiable from the surviving record examined.

7 The technical model

Represent a measured benchmark result as

$$Y = F(R, D, I, A, U),$$

where:

- R : recorded recipe—scenario, data, prompt construction, decoding, scorer;
- D : deployment—weights, model/tokenizer revisions, template, dtype, quantization;
- I : execution instrument—harness, engine, packages, kernels, hardware, topology, batching;
- A : artifact interpretation—schema, migration, canonicalization, carried-forward outputs;
- U : residual stochastic and numerical influences.

The public record exposes only a projection of these variables. Reproduction is identifiable at tolerance ϵ when all materially compatible configurations produce equivalent results under a declared output-, metric-, or conclusion-level criterion.

Recommended outcome classes:

1. Recipe mismatch
2. Deployment mismatch
3. Execution-instrument mismatch
4. Artifact-interpretation mismatch
5. Residual disagreement under controlled equivalence
6. Historically non-identifiable reproduction
7. Output-level reproduction
8. Metric-level reproduction
9. Robust conclusion despite non-identical outputs

7.1 Orthogonal targets of reproduction

The mismatch taxonomy classifies the source of disagreement. The paper should also state which object it aims to reproduce:

1. **Artifact reconstruction:** recompute published aggregates from the released outputs and metric records.
2. **Procedural reproduction:** rerun the recipe under a pinned or declared equivalent execution instrument.
3. **Claim-level robustness:** test whether rankings or improvement claims survive plausible substrate variation.

The released official artifact remains the ground truth for the historical score; local reruns test procedure stability and identifiability rather than replacing that record.

7.2 Result-store acceptance gate

A store is publication-ready only when all of the following are recorded: run-artifact freshness, comparison freshness, report freshness, frozen public target and run spec, model/tokenizer/deployment revisions, code commit, container/environment identity, raw outputs, and content hashes. A recent report timestamp alone is insufficient.

8 EvalAudit as the systems contribution

EvalAudit should be described as an evaluation auditing and experimental-control system, not merely a HELM wrapper.

8.1 Evaluation capsules

Each run should preserve:

- frozen benchmark specification, harness commit, and container digest;
- dataset and scorer versions;
- model and tokenizer repository revisions;
- chat-template text or hash and tokenized-prompt hash;
- dtype, quantization, engine, attention implementation, and generation configuration;
- batching, cache, tensor-parallel, numerical, GPU, CUDA, and driver settings;
- raw requests, token IDs, responses, metrics, and artifact-transformation history.

8.2 Layered differential comparison

EvalAudit should identify the earliest divergence layer:

recipe $\rightarrow$ prompt $\rightarrow$ tokens $\rightarrow$ generation $\rightarrow$ text $\rightarrow$ metric.

8.3 Controlled replay

The system should support counterfactual variation of one component while freezing the others: revision, tokenizer, template, special tokens, dtype, engine, harness era, and—where feasible—hardware-sensitive settings.

8.4 Structured reports

Reports should separate coverage and failure classes rather than emitting one undifferentiated reproduction rate.

9 The blocking empirical result

The highest-priority experiment is to move the OLMo candidate from the deployment probe into the real HELM path.

1. Freeze discovery instances and candidate configurations.
2. Determine whether the winning candidate requires probe-only `add_special_tokens=False` behavior.
3. Implement required behavior through the actual tokenizer/client path.
4. Execute the exact public `run_spec.json` through normal HELM.
5. Generate ordinary HELM artifacts.
6. Compare through the normal pair-report pipeline.
7. Evaluate on held-out instances and preferably another benchmark.

8. Preserve the full candidate surface, not only the winner.

A success supports a concrete attribution-and-closure claim. A failure remains publishable: it demonstrates that output-matching probes do not automatically translate into faithful benchmark execution.

10 Minimum empirical package

Study	Models	Benchmarks	Purpose
OLMo ordinary-path confirmation	2	2	Dense plus MoE; short deterministic plus IFEval; naive, exact recipe, reconstructed deployment.
OLMo bounded ablation	1–2	1–2	Dtype, template/generation prompt, and special-token behavior.
Repeatability baseline	1	1	Three same-machine runs; cross-machine only if operational quickly.
Historical case	1	1–2	Artifact/source proof; optionally two proxy-era executions.
Qwen demonstration	2	2–3	Compare model-generation delta with variation under one alternative plausible substrate.

Outcome metrics should include exact tokens, normalized text, first divergence position, common-prefix length, task-specific correctness or IFEval compliance, aggregate score difference, and ranking stability.

11 Small-model application

11.1 Scientific question

The future-facing question is:

Is the new or fine-tuned model genuinely better, or is the measured difference an artifact of the evaluation configuration?

For a model pair, compare

$$\Delta_{\text{model}} = Y_{\text{new}} - Y_{\text{old}}$$

with

$$\Delta_{\text{substrate}} = \max_c Y_{m,c} - \min_c Y_{m,c}.$$

A robust progress claim should exceed or remain stable across the tested substrate variation.

11.2 Qwen scope

Within the internship, use at most one carefully matched pair, exact model and tokenizer revisions, two or three benchmarks that discriminate within the small-model cohort, and a pinned EvalAudit environment. Separate native model templates from standardized evaluation semantics, and do not mix thinking/reasoning modes with direct-answer modes without a matched compute or token budget.

The purpose is to demonstrate EvalAudit’s relevance, not to establish a comprehensive Qwen leaderboard.

11.3 Fable-distilled models

Fable-tuned models become a headline result only if the following already exist:

- exact parent and teacher checkpoints;
- teacher prompts and sampling settings;
- training-data manifest, filters, deduplication, and contamination analysis;
- complete training hyperparameters, seeds, and checkpoint-selection rule;
- held-out and unrelated-task evaluation partitions.

Otherwise, include them as exploratory future work. Adding incomplete synthetic-data experiments would create more reviewer risk than value during the final weeks.

12 Cost and hardware

The paper should not describe hardware simply as an unmitigable source of non-reproducibility. Some historical hardware cannot be recreated, but it can be recorded, varied, bounded, and tested for conclusion robustness.

Distinguish:

1. **Exact replay:** same recorded instrument is available.
2. **Equivalent replay:** a different instrument produces outputs or metrics within a declared tolerance.
3. **Robust conclusion:** exact outputs differ, but the scientific conclusion or ranking is stable across tested instruments.

For current small models, report GPU-hours, peak memory, throughput, latency, and storage/audit overhead. This makes the paper relevant to local-model development and exposes whether a model’s gain is worth its resource cost.

13 Claims and language discipline

13.1 Safe claims

- A benchmark recipe does not fully specify the experiment.
- Exact-spec replay prevents documented recipe-reconstruction errors.

- Unrecorded tokenizer, template, dtype, and serving choices can materially change outputs in the studied cases.
- Some missing provenance is experimentally recoverable; some historical provenance is not uniquely recoverable.
- EvalAudit provides infrastructure to preserve and diagnose these layers.

13.2 Claims to avoid until further experiments

- Most HELM results are reproducible.
- Float32 was definitively the original TogetherAI configuration.
- Precision is the dominant hidden variable in general.
- Residual gaps are small or almost always attributable.
- Qwen 3.x/3.5 definitively advances over Qwen 2.5 across the capability spectrum.
- Fable fine-tuning reliably improves small models.

Use “highest-agreement tested candidate” rather than “the original deployment,” and “consistent with” rather than “proved” when multiple latent configurations remain possible.

14 Three-week execution plan

14.1 Days 1–3: freeze and draft

- Freeze thesis, contribution list, model/benchmark roster, and denominator.
- Build a run-status table separating probe, smoke, exact-path, stale, fresh, and completed results.
- Regenerate all non-GPU provenance analyses.
- Draft introduction, EEE relationship, technical model, EvalAudit system, and historical case.

At this point the manuscript should already be viable as a position/experience paper.

14.2 Days 4–8: OLMo confirmation

Resolve the request-path blocker and attempt one dense and one MoE model through ordinary HELM on two benchmarks. Regenerate reports with current code.

Decision gate If no ordinary-path confirmation is complete by the end of this block, stop broadening experiments. Preserve the failure as a result and finish the position/systems paper.

14.3 Days 9–12: bounded ablation and repeatability

Run a small staged or $2 \times 2 \times 2$ design over dtype, template/generation prompt, and special-token handling. Add three repeats of the final configuration. Prefer held-out confirmation over more factors.

14.4 Days 13–15: historical evidence and artifact

Regenerate hashes, source compatibility, and the historical timeline. Run a proxy-era comparison only if the path is already operational. Package immutable evidence and figure scripts.

14.5 Days 16–18: optional Qwen demonstration

Proceed only after the paper’s core empirical and artifact milestones are complete. Use one matched pair, two or three benchmarks, and one alternative plausible execution configuration.

14.6 Final days: freeze

Stop experiments, audit every claim against stored evidence, complete limitations, regenerate every figure from scripts, and archive the paper artifact.

15 Recommended display items

The highest-value figures and tables are:

1. Layered measurement-instrument diagram.
2. EEE-to-follow-on paper relationship diagram.
3. Corpus eligibility and completion funnel.
4. OLMo agreement waterfall: naive, exact recipe, reconstructed deployment.
5. OLMo bounded-ablation table or heatmap.
6. First-divergent-token examples.
7. Historical HELM source/artifact timeline.
8. Provenance field table: recorded, missing, material, recoverable, recommended.
9. Optional Qwen model delta versus substrate variation.
10. Capability-cost frontier for the small-model demonstration.

16 Submission-readiness criteria

The hybrid paper is ready when:

- inherited EEE results and internship contributions are explicitly separated;
- every result is labeled probe, smoke, exact-path, historical inference, or completed comparison;
- all cited tables are regenerated from current code and a frozen denominator;
- the ordinary-path OLMo outcome is reported honestly, whether success or failure;
- historical inferences are labeled as source-compatibility inferences;
- raw evidence, manifests, revisions, environments, and figure scripts are archived;

- the paper does not depend on unfinished Qwen or Fable results.

A broader empirical paper additionally requires held-out configuration validation, larger benchmark samples, controlled ablations, local-local and cross-machine baselines, completed modern-family breadth, and statistical uncertainty.

17 Decision after the collaborative review

The immediate manuscript should remain a hybrid technical position/systems paper. Its minimum empirical center is one ordinary-path OLMo confirmation attempt and the RedPajama/classic forensic contrast. A bounded Qwen comparison is valuable only after the core artifacts are secured; Fable-distilled models remain a follow-on unless their training and data provenance are already complete. Failure of the ordinary-path OLMo translation is itself publishable evidence if documented as a layered diagnosis rather than hidden as an incomplete result.

18 Consensus state

There is consensus on the paper’s story: benchmark reproduction is a layered system-identification problem; EEE detects discrepancies, while EvalAudit studies attribution, possible closure, and identifiability; OLMo is the partially recoverable modern case; classic HELM supplies proxy-era and non-identifiable historical cases; and a narrow Qwen study is optional validation rather than a dependency. The remaining disagreements are evidentiary labels and artifact-preservation requirements, not the scientific thesis.

19 Final recommendation

The highest expected-value strategy is:

Write the position and system paper now; use one end-to-end OLMo study to raise its technical level; add a narrow Qwen demonstration only if the core work finishes early; reserve the full small-model and Fable study for the expanded follow-on paper.

This plan preserves a publishable output under failure, directs scarce GPU time toward the single experiment that changes the paper’s evidentiary status, and leaves a coherent path to the higher-value small-model measurement paper.